

2026 - Group 1 Preliminary Exam - General Science

Answer Key | Q1-Q184, 13 questions

Q.No	Question	Correct Answer	Syllabus	Explanation
1	Assertion [A] : Databases of the Digital Herbarium simplify taxonomical references. Reason [R] : Databased images cannot help in confirming a specimen's historical collection data or location. (A) [A] is false, [R] is true (B) [A] is true, but [R] is false (C) Both [A] and [R] are true; and [R] is the correct explanation of [A] (D) Both [A] and [R] are true; but [R] is not the correct explanation of [A]	B) [A] is true, but [R] is false	Unit I - General Science	A Digital Herbarium is an online, searchable database of digitized (scanned/photographed) preserved plant specimens. By making thousands of specimens instantly searchable and comparable, it genuinely simplifies taxonomic identification and cross-referencing, so [A] is true. However [R] is false: digitized specimen records are usually paired with transcribed label metadata (collector name, date, locality/GPS coordinates), and this metadata is exactly what helps researchers confirm a specimen's historical collection data and location -- the opposite of what [R] claims. Hence the correct choice is (B). Source: Class 11 Botany, Unit 5 "Taxonomy and Systematic Botany"
2	Match the following : (a) Minamata disease (b) Itai-Itai disease (c) Black foot disease (d) Blue baby syndrome 1. Nitrate 2. Mercury 3. Cadmium 4. Arsenic (A) (a) 4 (b) 3 (c) 2 (d) 1 (B) (a) 2 (b) 3 (c) 4 (d) 1 (C) (a) 3 (b) 1 (c) 2 (d) 4 (D) (a) 1 (b) 2 (c) 3 (d) 4	B) (a) 2 (b) 3 (c) 4 (d) 1 [Minamata disease-Mercury, Itai-Itai disease-Cadmium, Black foot disease-Arsenic, Blue baby syndrome-Nitrate]	Unit I - General Science	Each of these is a classic disease caused by a specific pollutant. Minamata disease (Japan, 1950s) was caused by methylmercury poisoning from industrial wastewater discharged into Minamata Bay, damaging the central nervous system. Itai-Itai ("it hurts-it hurts") disease, also in Japan, was caused by chronic cadmium poisoning from mine wastewater, leading to painful bone and kidney damage. Black foot disease, seen in Taiwan, results from long-term consumption of arsenic-contaminated groundwater, causing peripheral vascular damage. Blue baby syndrome (methemoglobinemia) in infants is caused by excess nitrate in drinking water, often from over-fertilised groundwater, which reduces the blood's oxygen-carrying capacity. So the correct match is (a)-Mercury, (b)-Cadmium, (c)-Arsenic, (d)-Nitrate, option (B).

				Source: Class 11 Chemistry Volume 2, Unit 15 "Environmental Chemistry" and Class 12 Botany, Unit 8 "Environmental Issues"
24	_____ accounts for 14% of National Greenhouse Gas emissions and consumes 18% of national electricity in India. (A) Agriculture Sector (B) Industry Sector (C) Service Sector (D) Technology	A) Agriculture Sector	Unit I - General Science	As per India's official greenhouse gas inventories submitted to the UNFCCC, the agriculture sector (mainly from livestock enteric fermentation, paddy cultivation, and fertiliser use) accounts for roughly 14% of India's total national GHG emissions, behind only the energy sector. Agriculture is also a major electricity consumer because of widespread, often subsidised or free, groundwater irrigation pump-sets, which together account for about 18% of the country's total electricity consumption. This dual burden of being both a significant emissions source and a large, often inefficient, electricity consumer makes agriculture an important focus area for India's climate mitigation and energy-efficiency policies. Hence the answer is (A) Agriculture Sector.
51	Choose the correct statement from the following. (i) The flavour in yogurt is due to acetaldehyde. (ii) Yogurt is produced by bacterial fermentation of milk. (iii) Penicillin is produced by bacteria. (iv) Spirulina is a non edible unicellular microorganism. (A) (i) and (ii) are correct (B) (iii) and (iv) are correct (C) (ii) and (iii) are correct (D) (i) and (iv) are correct	A) (i) and (ii) are correct	Unit I - General Science	Yogurt is made by fermenting milk with bacterial starter cultures (mainly Lactobacillus bulgaricus and Streptococcus thermophilus), and its characteristic tangy flavour comes largely from acetaldehyde and other compounds these bacteria produce during fermentation, so statements (i) and (ii) are both correct facts of dairy microbiology. Penicillin, however, is produced by a mould/fungus (Penicillium notatum/chrysogenum), not by bacteria, making (iii) false. Spirulina is actually an edible, protein-rich cyanobacterium sold as a nutritional supplement, and it is filamentous rather than strictly unicellular, so (iv) is also false. Hence only (i) and (ii) are correct. Source: Class 12 Zoology, Unit 9 "Microbes in Human Welfare"
52	The Indian Institute of Technology - Madras has developed a technology called AMRIT. This technology is	B) the removal of arsenic and heavy metals from water	Unit I - General Science	AMRIT (Arsenic and Metal Removal by Indian Technology) is a water-purification technology developed by IIT Madras that uses a nano-scale iron-based adsorbent to remove

	integrated into the "Jal Jeevan Mission" to provide a highly effective solution for (A) the desalination of sea water (B) the removal of arsenic and heavy metals from water using nano technology (C) the water softening process (D) increasing the ground-water level	using nano technology		arsenic and other toxic heavy metals from drinking water at low cost, without requiring electricity. Because groundwater arsenic and heavy-metal contamination is a serious public-health problem in many parts of India, this technology has been integrated into the Jal Jeevan Mission's efforts to supply safe piped drinking water to every rural household, making the removal of arsenic and heavy metals the correct answer rather than desalination, softening, or raising the water table.
99	Choose the right matches in the following pairs : (1) Commensalism - One population benefits but second one is neither benefited nor affected. (2) Mutualism - Two populations living together but neither affects nor benefits. (3) Amensalism - One population prey on another at all times. (4) Competition - Populations are not affecting directly but fight for resources. (A) (2) and (3) are correct (B) (1) and (3) are correct (C) (1) and (4) are correct (D) (3) and (4) are correct	C) (1) and (4) are correct	Unit I - General Science	In the study of interspecific interactions in ecology, Commensalism is correctly defined as a relationship where one population benefits while the other is neither helped nor harmed - matching pair (1) exactly. Competition is correctly described as populations not directly affecting one another physically but competing indirectly for shared limited resources such as food, light or space - matching pair (4). The description given for pair (2) ('neither affects nor benefits') actually describes Neutralism, not true Mutualism (where both partners benefit), and the description for pair (3) ('one population preys on another at all times') describes Predation, not Amensalism (where one is harmed and the other is unaffected). Since only (1) and (4) are correctly matched to their ecological terms, option (C) is correct. Source: Class 12 Botany, Unit 6 "Principles of Ecology"
100	Assertion and Reason Type. Assertion [A] : Mendel's law of independent assortment the genes for each character separate and enter the gametes independently of the genes of the others characters. Reason [R] : Closely located genes assort independently. (A) [A] is correct but [R] is not correct	A) [A] is correct but [R] is not correct	Unit I - General Science	Mendel's Law of Independent Assortment correctly states that during gamete formation, alleles of different genes (controlling different characters) segregate and get distributed into gametes independently of one another, which is exactly what Assertion [A] describes, and is a well-established genetic principle observed for genes located on different chromosomes (or far apart on the same chromosome). However, Reason [R] is factually incorrect: genes that are closely located on the same

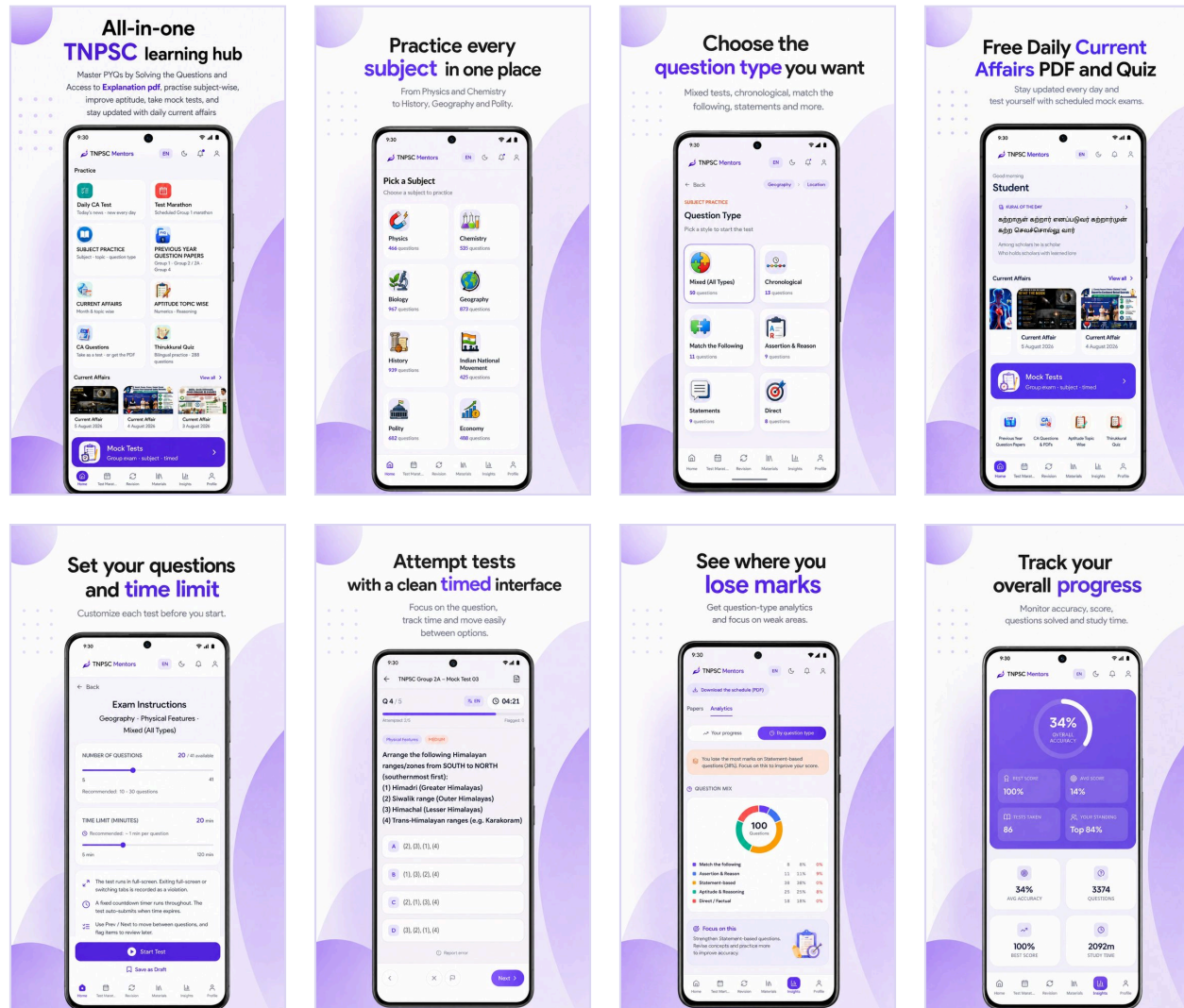
	(B) [A] is not correct but [R] is correct (C) Both [A] and [R] are correct and [R] is the correct explanation of [A] (D) Both [A] and [R] are correct but [R] is not the correct explanation of [A]			chromosome are said to be 'linked' and tend to be inherited together rather than assorting independently, because there is little chance of a crossing-over event occurring between them; independent assortment applies best to genes on different chromosomes or far apart, not closely located ones. Since [A] is true and [R] is false, option (A) is correct. Source: Class 12 Botany, Unit 2 "Classical Genetics" and Class 12 Botany, Unit 3 "Chromosomal Basis of Inheritance"
103	Which of the following is incorrectly paired? (A) IPCC - International Panel on Climate Change (B) CPC - Climate Prediction Centre (C) NOAA - National Oceanic and Atmospheric Administration (D) UNEP - United Nations Environment Programme	A) IPCC - International Panel on Climate Change	Unit I - General Science	IPCC actually stands for the 'Intergovernmental Panel on Climate Change', the UN body (established in 1988 by WMO and UNEP) that assesses climate science and is composed of representatives nominated by member governments - not an 'International Panel'. The other pairings are correctly expanded: CPC is the USA's Climate Prediction Centre, NOAA is the National Oceanic and Atmospheric Administration, and UNEP is the United Nations Environment Programme. Because option (A) misstates the full form of IPCC, it is the incorrectly paired option.
150	What is the major environmental advantage in using Chitosan encapsulated nano-urea over conventional urea? (A) It helps in maintenance of soil pH. (B) It provides drought tolerance to the crops. (C) It drastically reduce ground water contamination caused by nitrate leaching. (D) It speeds up the green house gas emissions from the soil.	C) It drastically reduce ground water contamination caused by nitrate leaching.	Unit I - General Science	Conventional granular urea is applied in bulk and has low nitrogen-use efficiency (typically only about 30-40% is absorbed by crops), with the surplus nitrogen leaching through the soil as nitrate and contaminating groundwater, besides volatilising as greenhouse gases. Chitosan-encapsulated nano-urea uses nanoparticle-sized urea coated in a biodegradable chitosan shell, which allows slow, targeted release of nitrogen that plants can absorb far more efficiently through foliar application, sharply cutting the surplus nitrogen available to leach into groundwater as nitrates. This environmental benefit - reduced nitrate-based groundwater contamination - is the key advantage, making option (C) correct; it does not particularly regulate soil pH, confer drought tolerance, or increase greenhouse gas emission (if anything it reduces

				such emissions), ruling out (A), (B) and (D).
151	Who was the first Indian Astronaut to visit the International space station? (A) Sunita Williams (B) Barry Wilmore (C) Nick Hague (D) Shubhanshu Shukla	D) Shubhanshu Shukla	Unit I - General Science	Group Captain Shubhanshu Shukla flew to the International Space Station in June 2025 as part of the Axiom Mission 4 (Ax-4), a commercial spaceflight organised by Axiom Space with SpaceX, becoming the first Indian astronaut ever to visit the ISS. He was only the second Indian in space overall, after Rakesh Sharma flew aboard the Soviet Soyuz T-11 in 1984, but Sharma went to the Salyut 7 space station, not the ISS. Sunita Williams and Barry Wilmore are American NASA astronauts (Williams is of Indian origin but holds US citizenship) who were on the same period's ISS crew, and Nick Hague is also a NASA astronaut, so none of them fits the description 'Indian Astronaut'. Hence option (D) is correct.
152	Which laboratory of NASA is used to study Quantum Physics in Microgravity? (A) Jet Propulsion laboratory (B) Cold atom laboratory (C) Space radiation laboratory (D) Johnson space center	B) Cold atom laboratory	Unit I - General Science	NASA's Cold Atom Laboratory (CAL), installed on the International Space Station in 2018, cools atoms to within a billionth of a degree above absolute zero, producing Bose-Einstein condensates whose quantum behaviour can be observed for far longer in microgravity than is possible under Earth's gravity. This makes CAL a unique facility for studying fundamental quantum physics, including atom interferometry and matter-wave phenomena. The Jet Propulsion Laboratory (which actually manages CAL) is NASA's centre for robotic planetary missions, the Space Radiation Laboratory studies cosmic radiation effects on biology, and Johnson Space Center is NASA's astronaut training and mission control hub, so none of these is the specific laboratory referred to in the question. Hence option (B) is correct.
162	Match the following National Sports Awards for the year 2024 with their respective recipients : Award (a) Arjuna Award (b) Dronacharya Award	B) (a)-2 (b)-1 (c)-4 (d)-3 i.e. Arjuna Award- Sucha Singh, Dronacharya Award- Subhash Rana, Khel Ratna Award-D. Gukesh, Maulana Abul	Unit I - General Science	At the National Sports Awards for 2024 (announced in January 2025), teenage chess Grandmaster D. Gukesh - who became World Chess Champion at just 18 - was among the recipients of the Major Dhyan Chand Khel Ratna Award, India's highest sporting honour, while Chandigarh University was conferred the Maulana Abul Kalam Azad (MAKA) Trophy for the best overall

	(c) Dhyan Chand Khel Ratna Award (d) Maulana Abul Kalam Azad Trophy Recipients 1. Subhash Rana 2. Sucha Singh 3. Chandigarh University 4. D. Gukesh (A) (a)-4 (b)-3 (c)-2 (d)-1 (B) (a)-2 (b)-1 (c)-4 (d)-3 (C) (a)-1 (b)-4 (c)-3 (d)-2 (D) (a)-3 (b)-2 (c)-1 (d)-4	Kalam Azad Trophy- Chandigarh University		performance by a university in inter-university sporting competitions, a title it has retained in multiple recent years. The Arjuna Award (for outstanding sporting performance) and the Dronacharya Award (for outstanding coaching) were given to other named individuals in the list; based on the fixed pairing of Khel Ratna with Gukesh and the MAKATrophy with Chandigarh University, only option (B) is internally consistent with these two well-established facts, making it correct.
184	The SHANTI Bill, 2025 is related to which one of the following sector? (A) Space technology (B) Nuclear energy (C) Renewable energy (D) Defence manufacturing	B) Nuclear energy	Unit I - General Science	As part of India's push to expand atomic power capacity (including Small Modular Reactors) under the Nuclear Energy Mission announced in the Union Budget, the government has been working on legislative changes to the Atomic Energy Act, 1962 and the Civil Liability for Nuclear Damage Act, 2010 to permit greater private and foreign participation in the nuclear power sector while addressing liability concerns. The SHANTI Bill, 2025 is associated with this nuclear-sector legal reform effort rather than with space technology, renewable energy, or defence manufacturing, which each have their own separate policy frameworks (e.g., the Space Policy for the space sector, or the Defence Acquisition Procedure for defence manufacturing). Given this, option (B) is the best-supported answer, though readers should note this is a very recent (2025) legislative development and should verify against the latest official notification. Hence option (B) is correct.

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